Dataset Documentation

European Housing Price Forecasting Project

Machine Learning and Deep Learning — CBS Spring 2026

Overview

This document describes every dataset used to construct the final modelling panel for our project. The project aims to forecast next-quarter residential house price growth across EU-27 countries using machine learning techniques, and to compare the predictive performance of models trained on data-driven versus literature-based country clusters.

The panel dataset is a country-quarter structure spanning approximately 2005 to 2025, covering up to 27 EU member states depending on data availability. Each row represents one country in one quarter. The ten datasets described below were selected to capture housing price dynamics from multiple complementary angles: demand-side economic conditions, supply-side constraints, financing costs, long-run structural factors, and the political environment.

Dataset	Source	Frequency	Role in Project
House Price Index (HPI)	Eurostat	Quarterly	Target variable
HICP Inflation	Eurostat	Monthly → Quarterly	Demand-side cost pressure
GDP Index	Eurostat	Quarterly	Economic activity
Unemployment Rate	Eurostat	Quarterly	Labour market conditions
Building Permits Index	Eurostat	Quarterly	Housing supply proxy
Old-age Dependency Ratio	Eurostat	Annual → Quarterly	Structural demographics
Population Density	Eurostat	Annual → Quarterly	Structural demographics
MIR Mortgage Rates	ECB API	Monthly → Quarterly	Cost of borrowing (eurozone)
Long-term Interest Rates	Eurostat	Quarterly	Sovereign yields (all EU-27)
Manifesto Project (MARPOR)	MARPOR	Per election → Quarterly	Parliament ideology score

1. House Price Index (HPI)

What it is

The House Price Index (HPI) is a quarterly index measuring the change in the price of residential properties purchased by households. The version we use is the nominal index with base year 2015 = 100, meaning a value of 110 indicates that prices are 10% higher than in 2015. It covers all types of residential dwellings — new and existing, houses and apartments — purchased on the open market.

This is the single most important variable in the project. Everything else we collect is collected because we believe it helps explain or predict this number.

Where we got it

Downloaded from the Eurostat Data Browser, dataset code prc_hpi_q. The specific filter applied was unit = I15_Q (index, base 2015 = 100, quarterly). The data is sourced by Eurostat from national statistical offices across EU member states and follows the internationally harmonised methodology defined in the Owner-Occupied Housing (OOH) Price Index framework.

Why we included it

The HPI serves two roles in the project. In its raw form it is used to compute the quarter-on-quarter percentage change (hpi_qoq_pct), which is the main feature representing current housing market momentum. Shifted forward by one quarter, it becomes the target variable (target_hpi_next_q) — what we are trying to predict. Every model in the project is trained to forecast this quantity.

We chose a QoQ growth rate rather than the index level itself for two reasons. First, growth rates are stationary, which is a requirement for most machine learning models that assume stable statistical properties over time. Second, QoQ growth captures short-term momentum, which is what a quarterly forecasting model needs to track.

What information it carries

- The current rate of house price inflation or deflation in each country
- Temporal momentum — markets that were rising last quarter tend to keep rising (captured via lags)
- Cross-country heterogeneity — some countries (e.g. Estonia, Hungary) experienced very rapid appreciation in the 2020-2022 period while others (e.g. Italy, Finland) were much more muted
- Cycle turning points — the index captures the 2008-2009 crash, the post-2015 recovery, the COVID-era boom, and the 2022-2023 correction driven by rate rises

2. Harmonised Index of Consumer Prices (HICP)

What it is

The Harmonised Index of Consumer Prices (HICP) measures the change in prices paid by households for a basket of goods and services. It is the standard measure of consumer price inflation across EU countries, designed to be comparable across member states. The index is harmonised — meaning the methodology and basket composition follow EU-wide rules — which makes it suitable for cross-country analysis. We use the total index (all items) with base year 2015 = 100, aggregated to quarterly frequency by averaging the three monthly observations within each quarter.

Where we got it

Downloaded from Eurostat, dataset code prc_hicp_midx. Filters applied: coicop = Total (all items), unit = Index, 2015=100. The raw data is monthly; we aggregate to quarterly using a simple mean of the three months in each quarter, which is consistent with standard practice in macroeconomic panel datasets.

Why we included it

Inflation affects housing prices through several channels simultaneously. On the demand side, high general inflation erodes real incomes and purchasing power, which can suppress demand for housing. On the supply side, construction cost inflation raises the price of new builds, feeding into both the new-build market and — through substitution effects — the existing property market. On the financial side, high inflation typically triggers interest rate increases by the ECB, which raises mortgage costs and cools demand.

Beyond these theoretical channels, housing is widely used as an inflation hedge — particularly by investors. When inflation rises, property becomes more attractive as a store of value, increasing investment demand and supporting prices. This creates a situation where the relationship between inflation and house prices can be positive or negative depending on which channel dominates, making it an empirically interesting variable.

What information it carries

- Current inflationary environment in each country
- Input cost pressure for housing construction
- Monetary policy signal — high HICP predicts tighter ECB policy which feeds into mortgage rates
- Real income effects — high inflation with stagnant wages reduces affordability
- In the model: QoQ change (hicp_qoq_pct) and its two lags capture how inflation dynamics evolve and feed into future price moves

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3. Gross Domestic Product (GDP) Index

What it is

The GDP variable is a chain-linked volume index of real Gross Domestic Product at market prices, expressed with base year 2005 = 100, seasonally and calendar adjusted. Chainlinking means the index accounts for changes in the composition of the economy over time, making it a more accurate measure of real economic growth than a fixed-basket index. Seasonal adjustment removes predictable quarterly patterns (e.g. higher economic activity in Q3 due to summer tourism), leaving the underlying trend and cyclical component.

Where we got it

Downloaded from Eurostat, dataset code namq_10_gdp (National Accounts quarterly). Filters: na_item = Gross domestic product at market prices, s_adj = Seasonally and calendar adjusted data, unit = Chain linked volumes, index 2005=100.

Why we included it

GDP growth is one of the most fundamental drivers of housing demand. When the economy is growing, household incomes rise, employment increases, and consumer confidence improves — all of which increase the number of households able and willing to purchase property, pushing prices up. Conversely, recessions reduce demand and can trigger sharp house price corrections, as was dramatically illustrated in several European countries during 2008-2012.

We use the index rather than the raw GDP level so that the variable is comparable across countries of very different sizes. The QoQ growth rate computed from this index (gdp_qoq_pct) measures the quarterly economic acceleration or deceleration, which is the relevant signal for short-term housing demand forecasting.

What information it carries

- Short-term economic momentum — the primary driver of housing demand cycles
- Business cycle position — whether the economy is expanding, contracting, or stagnating
- Income and employment expectations — GDP growth correlates strongly with wage growth and job creation
- The 2008-2009 and 2020 crashes are clearly visible as large negative QoQ observations, allowing the model to learn how housing markets respond to economic shocks

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4. Unemployment Rate

What it is

The unemployment rate measures the percentage of the labour force (age 15-74) that is unemployed and actively seeking work. The version we use is seasonally adjusted, total (all sexes), for the age group 15 to 74 years, expressed as a percentage of the labour force. This is the standard ILO-definition unemployment rate used in macroeconomic analysis across Europe.

Where we got it

Downloaded from Eurostat, dataset code une_rt_q (Unemployment by sex and age — quarterly). Filters: sex = Total, age = From 15 to 74 years, unit = Percentage of population in the labour force, s_adj = Seasonally adjusted data, not calendar adjusted data.

Why we included it

Unemployment affects housing markets through two main channels. The income channel is direct: households that lose their jobs lose their ability to service mortgage debt or pay rent, which both reduces demand for owner-occupied housing and can force distressed sales, adding supply and depressing prices. The confidence channel is subtler: even households that remain employed may postpone major purchases like homes when unemployment around them is high, due to the perceived risk of future job loss.

Unlike GDP, which is a flow measure of economic output, unemployment is a stock measure of labour market slack. The two are correlated but not identical — unemployment tends to lag GDP because firms are slow to hire and fire. This lagged relationship means unemployment adds information beyond what GDP already captures, particularly about household-level financial stress.

In the model we include both the level (unemployment_rate) and the quarter-on-quarter change (unemployment_qoq_change) to capture both the current labour market state and its direction of travel. A country with 8% unemployment but improving (falling) is in a very different position from one with 8% unemployment and rising.

What information it carries

- Labour market health and household financial resilience
- Mortgage default risk and forced supply
- Consumer confidence and willingness to commit to large purchases
- Directional signal: rising unemployment predicts price pressure even before it materialises

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5. Building Permits Index

What it is

The building permits variable is a quarterly index (base year 2021 = 100) measuring the number of residential building permits issued for dwellings, specifically covering residential buildings excluding communal residences (i.e. ordinary housing, not student dormitories or care homes). A building permit is a regulatory authorisation to begin construction — it is a leading indicator of future housing supply because it precedes actual construction by typically 6-18 months.

Where we got it

Downloaded from Eurostat, dataset code sts_cobp_q (Building permits — quarterly). Filters: indic_bt = Building permits - number of dwellings, cpa2_1 = Residential buildings, except residences for communities, unit = Index, 2021=100.

Why we included it

Housing prices are determined by both demand and supply. The other variables in our dataset are mostly demand-side (income, employment, borrowing costs). Building permits are the most important supply-side variable available at quarterly frequency across all EU countries.

When permits rise sharply, it signals that a large increase in housing supply is coming, which should eventually moderate price growth. When permits fall — as happened across much of Europe after 2022 in response to rising construction costs and financing difficulties — it signals that supply will tighten in the future, supporting prices even if demand is weak.

The forward-looking nature of permits is analytically useful: because they precede actual completions, a spike in permits this quarter partly predicts housing supply conditions two to four quarters from now. Our lag structure captures some of this dynamic.

What information it carries

- Current pipeline of new housing supply
- Developer and investor confidence in the housing market
- Supply-side constraints that prevent price adjustments
- Cross-country heterogeneity in planning and construction capacity

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6. Old-age Dependency Ratio

What it is

The old-age dependency ratio measures the number of people aged 65 and over per 100 people of working age (20-64). A ratio of 30 means there are 30 elderly people for every 100 working-age people. The data is annual; we expand it to quarterly frequency by repeating each year's value across all four quarters of that year, which is appropriate given the slow pace of demographic change.

Where we got it

Downloaded from Eurostat, dataset code demo_pjanind (Population and demographic projections, age dependency ratios). The data is published annually as part of Eurostat's demographic indicators.

Why we included it

Demographics are one of the deepest structural drivers of housing markets. Ageing populations affect housing demand in complex, non-linear ways. On the demand side, older households tend to downsize, increasing supply of larger family homes while increasing demand for smaller units and retirement housing. On the fiscal side, countries with high old-age dependency ratios face greater pressure on public finances, which can lead to higher taxes and reduced housing subsidies.

Crucially, this variable is used primarily in the clustering component of the project rather than directly in the predictive models. Countries with similar demographic profiles may share structural housing market characteristics — for instance, Eastern European countries have ageing and often shrinking populations, which creates very different long-run housing dynamics compared to countries with younger, growing populations.

What information it carries

- Long-run structural demand trajectory for housing
- Fiscal pressure that may affect housing policy
- Clustering signal: separates demographically young from demographically old markets
- Useful for explaining persistent cross-country differences in price dynamics that shorter-term economic variables cannot capture

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7. Population Density

What it is

Population density measures the number of inhabitants per square kilometre of land area. Like the old-age dependency ratio, this is an annual variable expanded to quarterly frequency by carry-forward. It captures the degree of urbanisation and land scarcity at the country level.

Where we got it

Downloaded from Eurostat, dataset code demo_r_d3dens (Population density by NUTS 3 region and country). We use the country-level aggregate.

Why we included it

Population density is a proxy for land scarcity and urbanisation pressure. In dense countries like the Netherlands (nearly 500 inhabitants per km²), there is limited undeveloped land available for new construction, creating a structural supply constraint that supports house prices over time. In sparse countries like Finland or Estonia, land scarcity is less binding and supply responses to demand can be larger.

Density also correlates with the extent to which the housing market is concentrated in a few large cities versus distributed across many smaller towns. Countries with extreme urban concentration (where most economic activity happens in one or two cities) tend to have more volatile housing markets than countries with more polycentric urban systems.

Like the old-age dependency ratio, this variable is primarily a clustering feature. It helps explain why structurally similar countries — for instance, the Benelux cluster of small, dense, highly urbanised economies — tend to have housing markets that behave similarly.

What information it carries

- Land scarcity and structural supply constraints

- Urbanisation intensity and the role of city-level dynamics
- Long-run price support in high-density, supply-constrained markets
- Clustering signal: useful for grouping countries by structural market characteristics

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8. MIR Mortgage Interest Rates (ECB)

What it is

The MIR (MFI Interest Rate Statistics) dataset published by the European Central Bank captures the actual interest rate that banks charge households on new mortgage loans, expressed as an annualised agreed rate (AAR). The series we extract is specifically for new business (new loans originated in that month, not the stock of outstanding loans), for house purchase, denominated in euros, covering credit and other institutions in the reporting sector.

The raw data is monthly. We aggregate to quarterly frequency by averaging the three monthly rates within each quarter, consistent with our treatment of HICP. The data is retrieved directly from the ECB API (data-api.ecb.europa.eu) rather than a pre-downloaded file.

Where we got it

Retrieved via the ECB Data Portal API, series key structure: MIR.M.{country}.B.A2C.AM.R.A.2250.EUR.N. The country dimension cycles through all eurozone members. The series starts in 2003 for most countries. Coverage is limited to eurozone members — the 20 EU countries that have adopted the euro. Non-euro EU countries (Denmark, Sweden, Poland, Czech Republic, Hungary, Bulgaria, Romania) do not have MIR data and will have NaN values for this variable.

Why we included it

Mortgage rates are arguably the single most powerful short-term driver of housing affordability. A household deciding whether to purchase a home must compare the monthly mortgage payment — which is directly determined by the interest rate — against its income. When rates rise from 1% to 4% on a typical mortgage, the monthly repayment increases by roughly 40%, immediately removing a large fraction of prospective buyers from the market and cooling demand.

This mechanism was dramatically illustrated across Europe in 2022-2023. As the ECB raised rates rapidly in response to post-pandemic inflation, mortgage rates across the eurozone rose from historic lows (below 1.5% in early 2022) to above 4% by late 2023. House price growth slowed sharply or reversed in most markets within two to three quarters of the rate increases — exactly the kind of dynamic our lagged model structure is designed to capture.

The reason we use MIR (actual mortgage rates paid by borrowers) rather than the ECB policy rate is that the transmission from policy rate to mortgage rates is imperfect and varies by country and time period. Banks may absorb some of the policy rate increase in their margins, or pass through more than 100% depending on competitive conditions. MIR captures the rate that actually matters for household decisions.

What information it carries

- The actual cost of mortgage financing for new borrowers in each country and quarter
- Monetary policy transmission — how ECB rate changes feed through to retail mortgage markets
- Credit market conditions — tighter lending standards often accompany rate rises
- Structural coverage note: NaN for non-eurozone countries is informative — these countries set their own monetary policy and have different interest rate environments, which is part of why the eurozone membership variable is analytically interesting
- In the model: included at level and with two lags, as mortgage rate changes typically take 2-4 quarters to fully affect transaction volumes and prices

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9. Long-term Government Bond Yields (Eurostat)

What it is

The long-term interest rate variable captures the yield on 10-year government bonds, measured as the EMU convergence criterion rate — the specific benchmark used to assess whether EU countries meet the interest rate convergence criterion for eurozone membership. It is expressed as a percentage per annum and is already available at quarterly frequency from Eurostat. Unlike the MIR mortgage rate which measures retail borrowing costs, this variable measures sovereign borrowing costs and serves as a proxy for long-run risk-free rates in each country's capital market.

Where we got it

Downloaded from Eurostat, dataset code irt_lt_mcby_q (Long-term interest rates for convergence purposes — quarterly data). The series goes back to the mid-1990s for most countries. The key advantage of this dataset over MIR is its coverage: all EU-27 countries have long-term rate data, including non-eurozone members, making it a useful complement to the MIR variable.

Why we included it

Long-term government bond yields serve as the foundation of the term structure of interest rates in each economy. They influence mortgage rates (particularly for fixed-rate mortgages, which are priced off long-term benchmarks), corporate borrowing costs (relevant for property developers), and the discount rate used in property valuations. A higher long-term yield generally makes housing less attractive relative to financial assets, cooling investment demand.

The combination of MIR and long-term yields gives us two complementary views of the interest rate environment. MIR captures the current cost of variable-rate or recently-originated mortgages for eurozone households. Long-term yields capture the broader capital market conditions and, for non-eurozone countries, provide the only interest rate signal we have. Together they give substantially better coverage and richness than either variable alone.

For non-eurozone EU members — particularly Central and Eastern European countries like Poland, Czech Republic, and Hungary — long-term yields are especially important because these countries set their own monetary policy. Their yield dynamics are driven by local inflation, fiscal positions, and exchange rate considerations rather than ECB decisions, creating interesting cross-sectional variation for the clustering analysis.

What information it carries

- The risk-free rate benchmark for property valuation and investment decisions
- Sovereign risk premium — higher yields signal perceived fiscal stress, as seen in Italy and Greece during the 2011-2012 European debt crisis
- Capital market conditions — yields falling to zero or below (as in Germany 2016-2020) created a structural push toward real assets including property
- For non-eurozone countries: the primary interest rate signal available
- Convergence dynamics: countries preparing to join the euro typically see their yields converge toward German levels, which affects their housing markets

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10. Manifesto Project Database (MARPOR)

What it is

The Manifesto Project (formally: the Comparative Manifesto Project, or MARPOR) is one of the most widely used datasets in political science. It systematically codes the content of party election manifestos using a detailed scheme of over 50 policy categories, producing quantitative measures of each party's ideological position on a range of dimensions. The dataset covers elections in democratic countries from 1945 to the present.

The key variable we extract is rile — the Right-Left position score. It is computed as the percentage of manifesto sentences falling into right-leaning policy categories minus the percentage falling into left-leaning categories. The resulting score ranges roughly from -100 (maximally left) to +100 (maximally right). Scores near zero indicate ideological centrists or parties with balanced left-right profiles.

We do not use the party-level rile scores directly. Instead, we construct a parliament-weighted ideology score for each country-election by computing the weighted average of all parties' rile scores, using each party's seat share (seats won divided by total seats) as the weight. This produces a single number representing the ideological centre of gravity of the legislature after each election.

Finally, we expand this to quarterly frequency using last-observation-carried-forward: each quarter's ideology score is the score from the most recent election, held constant until the next election occurs. This reflects the assumption that the policy environment created by an election persists between elections.

Where we got it

Downloaded from the Manifesto Project website (manifesto-project.wzb.eu) as the main dataset (mpd_data.csv). The dataset is freely available for academic use after registration. The version used is the 2025a release. Country names in MARPOR do not exactly match Eurostat conventions — for instance, MARPOR uses 'Czech Republic' while Eurostat uses 'Czechia', and MARPOR uses 'Greece' while Eurostat codes Greece as 'EL'. We built an explicit country name mapping to handle these inconsistencies.

Why we included it

Housing markets are profoundly shaped by policy. Rent regulation, property taxes, mortgage interest deductibility, zoning laws, social housing construction, and capital gains treatment all vary substantially across countries and change over time as governments change. A left-wing government tends to favour rent controls, social housing expansion, and restrictions on investment purchases — all of which affect supply, demand, and price dynamics in specific ways. A right-wing government tends to favour owner-occupation incentives, planning deregulation, and property rights — which also affect market dynamics but in different directions.

Rather than trying to capture every specific policy, the parliament_rile score provides a summary measure of the ideological direction of the government, which predicts the general orientation of housing policy. This is a methodologically defensible approach widely used in comparative political economy research.

It is important to note a limitation: MARPOR provides data on all parties that contested elections, not specifically on the governing coalition. We use the full parliament rather than just government parties because government membership data is not included in the standard MARPOR release, and coalition composition changes are complex. The full parliament score is a reasonable proxy — coalition governments in most EU countries tend to reflect the overall balance of parliament reasonably well — but this is an acknowledged limitation that we discuss in the report.

What information it carries

- The ideological orientation of the legislature, summarising expected direction of housing policy
- Political cycles — elections shift the score as governments change
- Persistence between elections — the score is stable within parliamentary terms, reflecting that policies take time to implement and unwind
- Cross-country structural differences — Southern European parliaments tend to score differently from Scandinavian ones on the rile scale, partly reflecting different institutional traditions around housing and welfare
- In the model: included at level and with two lags, since policy effects on housing markets typically materialise with a delay of several quarters

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Dataset Integration Summary

How the datasets fit together

The ten datasets combine to produce a panel with the following structure: each row represents one EU country in one quarter. The columns fall into five conceptual groups:

Group	Variables	Primary Use
Housing market	hpi, hpi_qoq_pct, target_hpi_next_q	Target + autoregressive feature
Macroeconomic demand	gdp_qoq_pct, hicp_qoq_pct, unemployment_rate, unemployment_qoq_change	Core predictors
Supply side	permits_qoq_pct	Supply pressure predictor
Financing costs	mortgage_rate_pct, lt_interest_rate	Affordability and monetary conditions
Structural / institutional	old_age_dependency_ratio, population_density, eurozone_member_timevarying, parliament_rile	Clustering + structural features

Missing data and coverage

Not all variables are available for all countries and all quarters. The main gaps are:

- mortgage_rate_pct: NaN for non-eurozone countries (DK, SE, PL, CZ, HU, BG, RO) throughout. This is structural rather than random missingness — these countries have their own monetary policy and mortgage markets not captured by the ECB MIR series. We retain the NaN rather than imputing, as the missingness pattern itself is informative alongside the eurozone membership variable.
- lt_interest_rate: Very few missing values (approximately 44 in model_data), primarily for Malta and Cyprus in the earliest quarters of the sample when these countries had not yet adopted the Maastricht reporting framework. Forward-filled.
- parliament_rile: NaN for quarters before the first election recorded in MARPOR for each country. Most EU countries have elections from the 1940s onwards in the dataset, so coverage is good from the start of our sample period (2005).
- old_age_dependency_ratio and population_density: Some missing values in early years for certain countries, forward-filled.

Frequency alignment

All datasets were harmonised to quarterly frequency before merging. Monthly series (HICP, MIR mortgage rates) were aggregated by taking the arithmetic mean of the three months within each quarter. Annual series (old-age dependency ratio, population density) were expanded by repeating each year's value across all four quarters. Election-level data (MARPOR) was expanded by carrying each election's ideology score forward to all subsequent quarters until the next election.

This approach is standard in macroeconomic panel data work. The key assumption — that annual or slow-moving variables can be treated as constant within a year — is appropriate for demographics and ideology, which genuinely do change slowly.

A note on the panel structure

The final modelling dataset contains approximately 1,700 rows after dropping observations with missing values in the core required variables (HPI growth rates and the target). This is deliberately conservative: we only require non-null values for the variables that are structurally present for all countries (the Eurostat core variables). Interest rate and ideology variables are allowed to have missing values in the model dataset — their missingness is handled either via imputation (forward-fill) or left as NaN for the model to handle.

With 26-27 countries and roughly 65-80 quarterly observations per country (spanning approximately 2005-2025), the panel is reasonably balanced. The variation is both cross-sectional (across countries at a given time) and temporal (within a single country over time), which is exactly the kind of two-dimensional variation that machine learning panel models can exploit.